



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION FEBRUARY 2025

Course: EC3247 (Cloud Architecture)
Lecturer: Mr. Izrulfizal Saufihamizal bin Ibrahim

Time: 8:00 am – 11:00am (3 hours)
Date: 27 May 2025

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 2 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. Explain the concept of "Rapid Elasticity" in cloud computing and illustrate your explanation with two real-world situations where elasticity plays an important role. (6 MARKS)
 - b. Assess the advantages and disadvantages of SaaS, PaaS, and IaaS for a medium-sized e-commerce company considering cloud migration, and justify which model would be most suitable. (12 MARKS)
 - c. Describe the principle of multi-tenancy in cloud computing and discuss its significance for cloud service providers and users. (4 MARKS)
 - d. Compare and contrast Public Cloud, Private Cloud, and Hybrid Cloud by highlighting their key differences and providing suitable examples for each. (3 MARKS)
2.
 - a. State **TWO(2)** motivation of adapting the cloud for web applications. (4 MARKS)
 - b. Justify the use of a dynamic infrastructure for a fast-growing startup with increasing customer traffic and expanding services. (6 MARKS)
 - c. Explain how to maintain cloud security. (6 MARKS)
 - d. A system has 20% of its workload as non-parallelizable (serial), while the rest can be parallelized. Using Amdahl's Law, calculate the speedup achieved with 4 processors. Then, evaluate the impact on speedup if the number of processors increases to 8. Support your answer with calculations and a clear explanation of the result. (9 MARKS)

3. a. Outline the standard steps involved in deploying an application to the cloud. **(6 MARKS)**
- b. Evaluate the benefits and drawbacks of using Virtual Machines (VMs) in cloud infrastructure management. **(6 MARKS)**
- c. Analyze the challenges cloud providers face in managing cost and performance in virtual infrastructures. **(4 MARKS)**
- d. Analyze the suitability of PaaS for a university planning to develop and deploy multiple academic applications for students and staff. **(9 MARKS)**
4. a. Describe the role and benefits of Virtualized Infrastructure Managers (VIMs) in cloud infrastructure management, highlighting three key features. **(7 MARKS)**
- b. Explain four attributes of integration platforms that are essential for effective cloud integration. **(8 MARKS)**
- c. Evaluate the impact of virtualization and open-source software on cloud computing adoption. Support your answer with at least two advantages and two challenges. **(10 MARKS)**

-END OF QUESTION PAPER-